Shengjie Zhu

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About me

My expertise lies in Spatial AI systems: designing customer-facing agentic front-ends grounded in a 3D-vision backend spanning camera calibration, localization, reconstruction, and Structure-from-Motion.

Work Experience

Aug, 2024 – now **Applied Scientist II, Ring AI, Amazon.**

1. Developed Ask Ring, a spatial-aware agentic system for Ring Cameras.
   – Led Intent Routing, Content Moderation, and primary contributor to Video Query Agent.
   – A robust LiDAR-Camera calibration method, friendly to inexperienced data-collection teams. [Publication-1]
   – Indoor & Outdoor Localization and Reconstruction. [Publication-2]
2. Optimized power- and memory-efficient Ring Detection models for embedded devices.

May – Aug, 2024 **Research Scientist Intern, BAIR, Google.**

Foundation depth model based Structure-from-Motion for Neural Rendering and AR/VR. [Publication-3]

2021 & 2022 **Applied Scientist Intern, Amazon Device AI.**

Improve the LiDAR generated depthmap quality with epipolar geometry. [Publication-5]

Education

2017 – 2024: **PhD, Computer Science**, *Michigan State University*, East Lansing, U.S.
Dissertation: Structure and Motion Estimation from Dense Depth and Correspondence
Advisor: Prof. Xiaoming Liu

2013 – 2017: **Bachelor of Engineering, Electrical and Electronics**, *Southeast University*, Nanjing, China.

Publications

(Under Review) **[1] Camera LiDAR Calibration in the Eye of LiDAR** [Website].
Shengjie Zhu, Feng Liu, Xiaoming Liu, Kah Kuen Fu, Chong Huang

(Under Review) **[2] On Self-Improving Structure-from-Motion: A Generalizable Multi-View Pose Estimator** [Website].
Shengjie Zhu, Ahmed Abdelkader, Yuyang Ji, Xiaoming Liu, Feng Liu, Kah Kuen Fu, Chong Huang

3DV'26 **[3] Marginalized Bundle Adjustment: Multi-View Camera Pose from MonoDepth** [PDF, Website].
Shengjie Zhu, Ahmed Abdelkader, Mark J. Matthews, Xiaoming Liu, and Wen-Sheng Chu

ECCV'24 **[4] Revisit Self-Supervision with Local Structure-from-Motion** [PDF, Website].
Shengjie Zhu, Xiaoming Liu

ECCV'24 **[5] Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry** [PDF, Website].
Shengjie Zhu*, Girish Chandar Ganesan*, Abhinav Kurmur, Xiaoming Liu

NeurIPS'23 **[6] Tame a Wild Camera: In-the-Wild Monocular Camera Calibration** [PDF, Website].
Shengjie Zhu, Abhinav Kurmur, Masa Hu, Xiaoming Liu

CVPR'23 **[7] LightedDepth: Video Depth Estimation in light of Limited Inference View Angles** [PDF, Website].
Shengjie Zhu, Xiaoming Liu

CVPR'23 **[8] PMatch: Paired Masked Image Modeling for Dense Geometric Matching** [PDF, Website].
Shengjie Zhu, Xiaoming Liu

CVPR'20 **[9] The Edge of Depth: Explicit Constraints between Segmentation and Depth** [PDF, Website].
Shengjie Zhu, Garrick Brazil, Xiaoming Liu

Computer Skills

Language **Python, C++, CUDA, Matlab, PyTorch, Tensorflow, CuPy, Numba**

Talk

Aug. 08, 2023 **3D Perception from Two Views**, Google Pixel Biometrics Seminar.

Feb. 05, 2024 **Structure-from-Motion Meets Self-supervised Learning**, CMU VASC Seminar. [Link]

Jul. 09, 2024 **Structure-from-Motion from Dense Learning**, Google Pixel Biometrics Seminar.